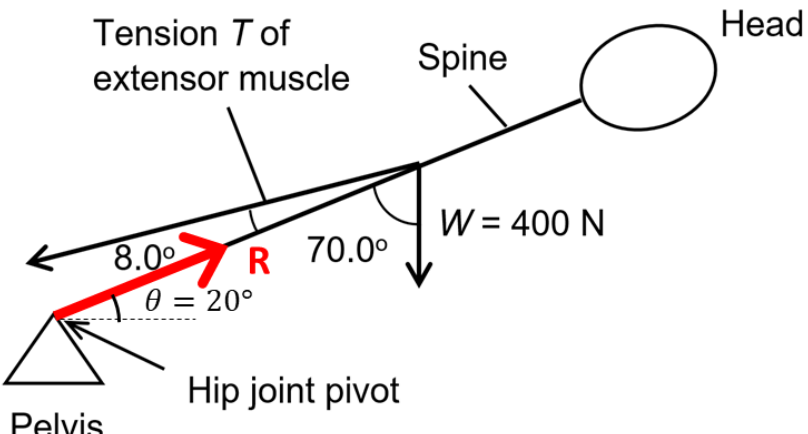
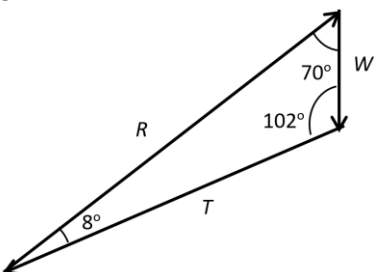


2ai	 Tension T of extensor muscle Spine Head $W = 400 \text{ N}$ 8.0° $\theta = 20^\circ$ R 70.0° Pelvis Hip joint pivot	B1
aii	As R, W and T intersect at a point, taking <u>moment about that point</u> would mean that there is <u>no net moment</u>. This would allow the person to be in rotational equilibrium.	B1
bi	Since the person is in translational equilibrium, there is <u>no net force on the body</u>. Hence, R, W and T must form a closed vector triangle.	B1
bii	$\sum F_y = 0: T \cos 78 + 400 = R \sin 20 \dots (1)$ $\sum F_x = 0: T \sin 78 = R \cos 20$ $T = \frac{R \cos 20}{\sin 78} \dots (2)$ Sub into (2) into (1) $\left(\frac{R \cos 20}{\sin 78} \right) \cos 78 + 400 = R \sin 20$ $R = 2811.3 \approx 2800 \text{ N}$ $T = \frac{(2811.3) \cos 20}{\sin 78} = 2700 \text{ N}$ OR  Using sine rule: $\frac{W}{\sin 8^\circ} = \frac{R}{\sin 102^\circ} = \frac{T}{\sin 70^\circ}$ $\frac{400}{\sin 8^\circ} = \frac{R}{\sin 102^\circ} = \frac{T}{\sin 70^\circ}$ Solving for R and T we get: $R = \underline{2.8 \times 10^3 \text{ N}}$ $T = \underline{2.7 \times 10^3 \text{ N}}$	C1 C1 A1 A1

c	When the spine is almost vertical, the <u>moment due to the weight about the pivot is reduced to a much smaller value</u>. For rotational equilibrium, the counteracting moment due to the <u>tension T in the extensor muscle is also small</u>. So T will also be small, and so the back muscle won't be strained.	B1 B1